

Field Reflection Journal II – Manipulatives

Course: Mathematics Methods Date of Observation: 4/28/2026 Grade Level Observed:
6th Grade

Description of the Lesson

The math lesson I observed was a 6th-grade class focused on statistics and data analysis, covering measures of central tendency such as mean, median, and mode, mean absolute deviation also known as MAD, and quartiles with interquartile range also known as IQR. These concepts aligned with the student notes captured in Images 1 and 2. Image 1 showed notes defining mean as the average of a data set, median as the middle number when arranged numerically, and mode as the most frequently occurring number, alongside a worked example using the data set 9, 9, 13, 15, 16, 21, and 26. Image 2 extended the lesson into MAD, described as the average distance each point is from the mean, and quartiles, defined as points that divide data into four equal parts, with IQR representing the range from the first to the third quartile. The primary manipulatives used were algebra tiles and number line strips, supplemented by printed histogram handouts like those in Images 3 and 4, which featured real-world frog jumping data from a Calaveras County competition comparing 2008 and 2009 results.

Teacher's Use of Manipulatives

The teacher began by activating prior knowledge, asking students what they already knew about averages before introducing formal vocabulary. Algebra tiles were then distributed to help students physically represent data values and model the concept that

finding the mean involves an evening out of values, connecting to the additive inverse property. For the MAD portion shown in Image 2, the teacher used number line strips where students placed data points and measured each point's distance from the mean using unit tiles, directly modeling the definition in the student notes. When transitioning to quartiles and IQR, color-coded data cards were introduced so students could physically sort and divide data into four equal groups. The teacher also used think-alouds throughout, narrating each step before students worked independently, and consistently referenced the inverse property of addition to bridge statistical and algebraic reasoning.

Student Interaction and Engagement

Students were enthusiastic when manipulatives were first introduced, though some off-task behavior appeared early. A few students stacked algebra tiles or built shapes instead of using them mathematically. The teacher responded calmly by restating the purpose of the tiles without singling out any student, which refocused the class without disrupting lesson flow. On-task behavior was strongest during the quartile card-sorting activity, where small groups arranged data cards to identify Q1, Q2, and Q3. Peer discussion was rich, with students explaining to each other why the median of the lower half identifies Q1, showing genuine conceptual understanding. When some students confused IQR with the total range, the teacher used the physical cards to point directly to the relevant data sections, making the distinction clear in a way that a verbal explanation alone may not have achieved.

Effectiveness of Manipulatives

The manipulatives were highly effective in making abstract concepts concrete. The number line strips during the MAD activity transformed a procedural calculation into a geometric idea by having students physically measure the distance of each data point from the mean. This helped students grasp why MAD is always non-negative, since distance cannot be negative, which aligned with the number line diagram shown in Image 2. The real-world frog jumping data from Images 3 and 4 added further effectiveness by giving students a genuine reason to engage with the statistics. Guided by the teacher, students compared the center, spread, and outliers of the 2008 and 2009 data sets, and through sorting the data cards they were able to observe that the 2009 results were more consistent, supporting the conclusion referenced in Image 3. Physical manipulation of the data produced a deeper analytical experience than reading a table alone would have.

Personal Reflection

This observation reinforced my belief that mathematics should not be taught as a purely procedural subject. Watching students sort data cards, measure distances on number lines, and use tiles to model the mean showed me how manipulatives make complex concepts accessible to more learners. Going forward, I plan to use color-coded data cards for teaching quartiles so students construct the concept through action before encountering formal notation. I also intend to anchor lessons in real-world data like the frog jumping sets in Images 3 and 4, since that context generated genuine curiosity and sustained engagement throughout the lesson.